Resume – Data Science & Machine Learning

João Pedro Mateus Souza

Software Engineer | **Data & AI** | **Machine Learning**

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PROFESSIONAL SUMMARY

Final-year Computer Science student and Software Engineer with professional experience in full-stack development, data analytics, machine learning, AI-powered automation, and software engineering. Experienced in building end-to-end platforms with **Python and Django**, integrating **REST APIs**, modeling relational databases, and implementing **LLM-based document analysis solutions**. Strong background in deep learning, computer vision, data pipelines, and business-process automation, with the ability to translate complex requirements into scalable technical products. Seeking full-time opportunities in **Software Engineering, Data, Machine Learning, or AI**.

CORE SKILLS

Data & AI
Machine Learning · Deep Learning · Computer Vision · Semantic Segmentation · Large Language Models (LLMs) · AI Agents · Data Analytics · KPI Design · Data Cleaning & Augmentation · Data Scraping · Data Normalization · Model Evaluation

Programming & Tools
Python · SQL · Bash · Docker · Git · REST APIs · Flask · Django · OpenStack · Linux · Automated Testing

Cloud & Infrastructure
AWS · Azure · Google Cloud · Virtual Machines · Containerization

Databases
MySQL · PostgreSQL · SQLite · ClickHouse

Methodologies
Agile (Scrum, Kanban) · Lean Six Sigma

PROFESSIONAL EXPERIENCE

Elevential Digital Solutions – Full-Stack Developer

2026 – Present

- Develop end-to-end web platforms using primarily **Python, Django, relational databases, and REST APIs** for clients across different business domains.
- Design and implement **real estate credit origination platforms**, supporting workflows from application intake and document processing to credit analysis and operational management.
- Model business processes, data structures, and system components to support the scalability, maintainability, and evolution of credit products.
- Integrate external APIs from notary and property registry services, automating document retrieval, validation, and real estate transaction workflows.
- Implement **LLM-based document analysis solutions** to extract, classify, validate, and interpret information from financial, legal, and property-related documents.
- Orchestrate **AI agents and LLM-assisted development workflows** to support software engineering, process automation, and solution delivery.
- Develop an academic conference management platform, participating in product discovery, requirements gathering, product modeling, system architecture, implementation, testing, and deployment.
- Collaborate directly with stakeholders throughout the full product lifecycle, translating business requirements into scalable technical solutions.

Nokia Corporation – Automation & Data Analytics Intern

2024 – 2026

- Designed and delivered **end-to-end automation and data analytics solutions** supporting strategic decision-making across telecom operations.
- Developed **interactive dashboards** and analytical tools to monitor operational KPIs and performance indicators.
- Built and maintained **data pipelines**, including data scraping, cleaning, transformation, and visualization.
- Modeled and optimized **relational databases** to support analytics and governance requirements.
- Developed **web-based analytics platforms** using Python (Flask), SQL, and REST APIs.
- Implemented containerized solutions using **Docker**, with deployments on **OpenStack virtualized environments**.
- Contributed to automation and monitoring solutions for **telecommunications infrastructure across Canada**.

PROFESSIONAL EXPERIENCE

Acceta Especialistas – Technical Consultant & Developer

2023 – 2024

- Worked in an **R&D and innovation consultancy**, supporting technology-driven projects and client proposals.
- Authored and automated **technical project descriptions and R&D reports**, improving productivity and consistency.
- Leveraged **Large Language Models (LLMs)** to support structured technical writing and ideation.
- Co-developed the company's first **internal project management and documentation platform** using **Django and MySQL**.
- Participated in client meetings, translating business requirements into technical solutions.

PROJECTS

NFC-e API – Domestic Purchase Data Management and Analytics

Python · REST APIs · Relational Databases · Docker · Data Normalization · Automated Testing

- Developed an API for extracting, organizing, and analyzing household purchase data from Brazilian electronic consumer invoices (**NFC-e**).
- Integrated the platform with **SEFAZ** services using serialized request processing and configurable request intervals to reduce rate limiting and temporary blocks.
- Implemented targeted retry and error-handling policies for HTTP 429, server errors, network failures, unavailable invoices, and CAPTCHA-protected pages.
- Created an assisted workflow that accepts invoice HTML retrieved through a browser or WebView when automated access requires CAPTCHA resolution.
- Designed a conservative product-normalization model that preserves original descriptions and links items to canonical records containing category, brand, unit, and reference quantity.
- Built idempotent command-line routines for dry-run simulation and incremental normalization, alongside automated tests using Python's `unittest` framework.

Semantic Image Segmentation with Convolutional Neural Networks

Python · Deep Learning · Computer Vision · CNNs · Data Augmentation · Model Evaluation

- Designed and trained **Convolutional Neural Networks** for semantic image segmentation and pixel-level classification.
- Built preprocessing and data-augmentation workflows to prepare image datasets for model training and experimentation.
- Evaluated segmentation performance through structured experiments and investigated the influence of model architecture, preprocessing, and training configurations.

ACADEMIC EXPERIENCE

Scientific Initiation – Computer Vision (UFF)

- Conducted research in **Deep Learning and Computer Vision**, with focus on **semantic image segmentation**.
- Designed and trained **Convolutional Neural Networks (CNNs)** for pixel-level classification tasks.
- Gained practical experience with data preprocessing, model evaluation, and experimentation workflows.

PET-Telecom – Tutorial Education Program (UFF)

- Developed projects using **Octave/MATLAB**.
- Co-authored an **educational article on Number Systems**.

EDUCATION

Bachelor of Science in Computer Science

Fluminense Federal University (UFF) – Niterói, RJ, Brazil

Final Year Student

CERTIFICATIONS & COURSES (LINKEDIN LEARNING)

- Data Science Foundations: Fundamentals
- Artificial Intelligence Foundations: Thinking Machines
- AI Foundations: Machine Learning & Deep Learning
- Introduction to Artificial Intelligence
- Learning Docker
- Process Improvement Foundations
- Developing Business Acumen
- Leading with a Growth Mindset

LANGUAGES

- English – Advanced
- Portuguese – Native
- French – Beginner